Gajula Yashaswi Sri Ram

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.73
Narayana JR College, SR Nagar, Hyderabad	Class XII (TSBIE)	2024	96.90%
Vinner's High School, Nirmal, Telangana	Class X (BSE Telangana)	2022	9.70%

PROJECTS

Multithreaded HTTP Server from Scratch

- Built an **HTTP server from sockets**, implementing request parsing, routing, and response generation **without any web framework**.

- Designed a **fixed size thread pool** with a **synchronized queue** to handle **concurrent connections**, ensuring **thread safe processing** and **logging**.

- Implemented **static file serving** with **MIME resolution**, **traversal protection**, and custom **HTTP error responses (403/404/405)**.

- **Stress tested** the server with **200 requests**, validating **stability** and **correctness under load**.

Chess Engine

- Built a complete **legal move generator** for all pieces including **castling, en passant, and pawn promotion**.

- Implemented **minimax with alpha-beta pruning** achieving **instant depth-3** and 3-4 second depth-4 full search.

- Designed board evaluation using **piece-square tables** for positional awareness and material scoring combined.

- Architected clean separation between game logic, search engine, and UI for **modular testing** and extensibility.

Custom Memory Allocator

- Implemented a custom heap allocator using a **doubly linked free list** with **block splitting** and **coalescing** of adjacent free blocks

- Managed memory from a single **pre-allocated pool**, avoiding repeated calls to system malloc/free for allocation requests

- Used a **first-fit search strategy** across the free list for fast O(n) block lookup during allocation

- Validated with **tests** covering allocation, deallocation, splitting, coalescing, and **edge cases such as out of memory and null-pointer frees**

8 bit Breadboard Computer

- Built an 8-bit computer from **74-series logic chips** following Ben Eater's architecture with ALU, registers, and control unit.

- Implemented **program counter, memory interface, clock, and bus arbitration** on breadboards using **fetch-decode-execute** cycle.

- Wired and **debugged** the complete system using a **logic analyzer** and verified instruction execution.

Digital Systems Lab, FSM Based Clock and Traffic Controller

- Designed a **digital clock with alarm** functionality and a **4-way traffic light controller** with pedestrian signals.

- Implemented timer, alarm logic, pedestrian crossing sequences, and **emergency vehicle override** using state transitions.

- Developed comprehensive **testbenches** with edge-case stimulus and verified system functionality through **waveform analysis**.

Battery Management System PCB (ESP32, isoSPI)

- Led PCB layout in **Altium Designer** for a Battery Management System board interfacing **ADBMS1818 12-channel battery monitor IC** with an **ESP32 over isoSPI**, as part of a **5-member hardware team**.

- Handled schematic to layout translation, component placement, and **design rule verification (DRC)**.

- Generated and validated gerbers, drill files, and BOM for a fabrication order **sent to a PCB manufacturer**.

SKILLS

- **Languages:** C++, C, Python, JavaScript, SQL
- **Tools & Platforms:** Git, GitHub, Linux, Bash, Docker, Jira, LaTeX
- **Concepts:** Data Structures & Algorithms, OOP, Memory Management, Computer Architecture, Multi-threading, RAG Pipelines, Embeddings, Vector Databases , Prompt Engineering
- **Coding Profiles**: LeetCode

RELEVANT COURSES

Computer Science: Data Structures & Algorithms, Machine Learning

Computer Architecture: Microprocessors & Computer Architecture, Digital Systems

Mathematics: Vector Calculus, Probability & Random Processes, Numerical Methods for Device Modeling

POSITIONS OF RESPONSIBILITY

Hospitality Coordinator, SPIC MACAY, IIT Hyderabad

- Coordinated hospitality and logistics across multiple teams for large scale cultural events hosted on campus

- Managed on ground execution during events, ensuring smooth guest and artist experience

- Worked closely with peers and event leads to plan resource allocation and scheduling ahead of each event

Technical Coordinator, Canteen Monitoring Committee, IIT Hyderabad

- Managed technical and digital logistics for the committee.

- Coordinated between students and administration to resolve service issues.